

Proposed Core Experimental Test

Phase-Angle Sweep for a Narrow Structural Response Basin

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Follow-up note for discussion with Dr Tan Sui, University of Surrey

Purpose

The purpose of this proposed test is to reduce the broader GEM idea to one simple experimental question:

Can a structured material under controlled cyclic excitation show a narrow, reproducible response optimum as the excitation phase is swept from 210° to 300°?

The aim is not to validate the wider framework in one step. The aim is only to determine whether a non-generic, phase-dependent structural optimum exists at all.

Core Test

Material platform: a single well-characterised ferroelectric benchmark material.

Excitation variable: controlled phase-angle sweep from 210° to 300°.

Sampling strategy: coarse sweep first, followed by finer sampling around any candidate basin.

Control logic: hold geometry, environment, amplitude range, and frequency range fixed within each test set.

Measurement Goal

The strongest candidate signature is not merely loop displacement, since simple shift can arise from ordinary asymmetries or instrumental effects. The more meaningful signal would be a narrow phase window in which the hysteretic response becomes measurably sharper, cleaner, and more reproducible.

Primary Measurable Outputs

Output	Why it matters
Hysteresis loop area	Captures overall loop narrowing or broadening
Switching slope	Tests whether the loop becomes sharper
Coercive width	Tracks spread of switching threshold
Branch roughness	Proxy for irregularity / mismatch
Cycle-to-cycle variance	Measures repeatability of response

Positive Signal

A narrow phase window showing a reproducible local optimum in loop organisation.

Steeper switching walls.

Reduced loop smearing.

Narrower coercive spread.

Smoother branch structure.

Lower cycle-to-cycle variance.

Negative / Null Result

Broad or weak phase dependence only.

Generic loop displacement with no local optimum in structural quality.

No clear narrowing, sharpening, or reproducibility gain beyond ordinary instrumental behaviour.

Minimal Experimental Question

When a ferroelectric benchmark material is driven under cyclic excitation with phase swept from 210° to 300°, does a narrow basin of hysteresis-loop sharpening appear?